Lines and Planes in Three Dimensions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: The vector equation of a line**

Write the vector equation of the line through the point $(2, -1, 3)$ with direction vector $(1, 2, -1)$.

$\mathbf{r} = (2, -1, 3) + t(1, 2, -1)$, where $t \in \mathbb{R}$.

2 **[Foundational] • Target: The vector equation of a line**

Find the vector equation of the line through the two points $A(1, 0, 2)$ and $B(4, 3, 1)$.

Direction vector: $\vec{AB} = (4-1,\ 3-0,\ 1-2) = (3, 3, -1)$. Vector equation: $\mathbf{r} = (1, 0, 2) + t(3, 3, -1)$.

3 **[Foundational] • Target: Determining if two lines intersect**

Determine whether the lines $\mathbf{r}_1 = (1, 2, 0) + s(1, -1, 2)$ and $\mathbf{r}_2 = (0, 3, 1) + t(2, 1, -1)$ intersect, and if so, find the point of intersection.

Equate components: $1 + s = 2t$, $2 - s = 3 + t$, $2s = 1 - t$. From the second: $s = -1 - t$. Substitute into the first: $1 + (-1 - t) = 2t$, so $-t = 2t$, giving $t = 0$, then $s = -1$. Check the third equation: $2(-1) = 1 - 0$? $-2 = 1$ — false. The lines do not intersect (they are skew, since they are not parallel either).

4 **[Foundational] • Target: The equation of a plane**

Find the Cartesian equation of the plane with normal vector $(2, -1, 3)$ passing through the point $(1, 2, -1)$.

The plane has the form $2x - y + 3z = d$. Substituting the point: $2(1) - 2 + 3(-1) = 2 - 2 - 3 = -3$. Equation: $2x - y + 3z = -3$.

5 **[Foundational] • Target: The equation of a plane**

Find the Cartesian equation of the plane containing the points $A(1, 0, 0)$, $B(0, 1, 0)$, and $C(0, 0, 1)$.

$\vec{AB} = (-1, 1, 0)$, $\vec{AC} = (-1, 0, 1)$. Normal $= \vec{AB} \times \vec{AC} = (1 \cdot 1 - 0 \cdot 0,\ 0 \cdot (-1) - (-1) \cdot 1,\ (-1) \cdot 0 - 1 \cdot (-1)) = (1, 1, 1)$. Plane: $x + y + z = d$; substituting A: $1 + 0 + 0 = 1$. Equation: $x + y + z = 1$.

6 **[Foundational] • Target: Angle between a line and a plane**

Find the angle between the line with direction $(1, 2, 2)$ and the plane $x + y + z = 5$, to the nearest degree.

Plane normal: $\mathbf{n} = (1, 1, 1)$. $\mathbf{d} \cdot \mathbf{n} = 1 + 2 + 2 = 5$. $|\mathbf{d}| = 3$, $|\mathbf{n}| = \sqrt{3}$.
$\sin\theta = |\mathbf{d} \cdot \mathbf{n}| / (|\mathbf{d}||\mathbf{n}|) = 5/(3\sqrt{3}) \approx 0.962$, so $\theta \approx 74^\circ$.

© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • **Target: Distance from a point to a plane**

Find the shortest distance from the point $(3, 1, -2)$ to the plane $2x - y + 2z = 6$, using $d = \frac{|ax_0 + by_0 + cz_0 - k|}{\sqrt{a^2 + b^2 + c^2}}$.

$d = |2(3) - 1(1) + 2(-2) - 6|/\sqrt{4 + 1 + 4} = |6 - 1 - 4 - 6|/3 = |-5|/3 = 5/3$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • **Target: Finding the angle between two planes**

Find the angle between the planes $x + 2y - z = 3$ and $2x - y + z = 5$, to the nearest degree.

Normals: $\mathbf{n}_1 = (1, 2, -1)$, $\mathbf{n}_2 = (2, -1, 1)$. $\mathbf{n}_1 \cdot \mathbf{n}_2 = 2 - 2 - 1 = -1$. $|\mathbf{n}_1| = \sqrt{6}$, $|\mathbf{n}_2| = \sqrt{6}$. $\cos\theta = |-1|/(\sqrt{6} \cdot \sqrt{6}) = 1/6$, so $\theta \approx 80^\circ$.

9 [Intermediate] • **Target: Finding the intersection line of two planes**

Find a vector equation for the line of intersection of the planes $x + y + z = 6$ and $x - y + 2z = 5$, using the cross product of the two normal vectors as the direction, and finding one point on both planes by inspection.

Direction: $\mathbf{n}_1 \times \mathbf{n}_2 = (1, 1, 1) \times (1, -1, 2) = (1 \cdot 2 - 1 \cdot (-1),\ 1 \cdot 1 - 1 \cdot 2,\ 1 \cdot (-1) - 1 \cdot 1) = (3, -1, -2)$. To find a point, try $z = 0$: $x + y = 6$ and $x - y = 5$, giving $x = 5.5$, $y = 0.5$. Point: $(5.5, 0.5, 0)$. Line: $\mathbf{r} = (5.5, 0.5, 0) + t(3, -1, -2)$.

10 [Intermediate] • **Target: Testing whether a line lies within a plane**

Determine whether the line $\mathbf{r} = (1, 2, 3) + t(2, -1, 1)$ lies entirely within the plane $x + 3y + z = 10$.

A line lies within a plane if (1) its direction vector is perpendicular to the plane's normal, and (2) a point on the line satisfies the plane's equation. Normal of the plane: $(1, 3, 1)$. Direction of line: $(2, -1, 1)$. Dot product: $1(2) + 3(-1) + 1(1) = 2 - 3 + 1 = 0$ ✓ (perpendicular to normal, so parallel to the plane). Check the point $(1, 2, 3)$: $1 + 3(2) + 3 = 1 + 6 + 3 = 10$ ✓ (satisfies the plane equation). Since both conditions hold, the line lies entirely within the plane.

11 [Intermediate] • **Target: Finding the point where a line meets a plane**

Find the point where the line $\mathbf{r} = (1, 0, 2) + t(1, 2, -1)$ intersects the plane $2x + y - z = 3$.

Substitute the parametric coordinates $x = 1 + t$, $y = 2t$, $z = 2 - t$ into the plane equation: $2(1 + t) + 2t - (2 - t) = 3$. Expand: $2 + 2t + 2t - 2 + t = 3$, so $5t = 3$, giving $t = 3/5$. Substituting back: $x = 1 + 3/5 = 8/5$, $y = 2(3/5) = 6/5$, $z = 2 - 3/5 = 7/5$. Point of intersection: $(8/5, 6/5, 7/5)$.

© ClipSAT — clipsat.org | All Rights Reserved

12 [Intermediate] • Target: Shortest distance between two skew lines

Two skew lines have direction vectors $\mathbf{d}_1 = (1, 0, 1)$ and $\mathbf{d}_2 = (0, 1, 1)$, and pass through the points $A(0, 0, 0)$ and $B(1, 1, 0)$ respectively. Find the shortest distance between the two lines, using $d = |\vec{AB} \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$.

$\mathbf{d}_1 \times \mathbf{d}_2 = (0 \cdot 1 - 1 \cdot 1,\ 1 \cdot 0 - 1 \cdot 1,\ 1 \cdot 1 - 0 \cdot 0) = (-1,\ -1, 1)$. $\vec{AB} = (1, 1, 0)$. $\vec{AB} \cdot (\mathbf{d}_1 \times \mathbf{d}_2) = 1(-1) + 1(-1) + 0(1) = -2$. $|\mathbf{d}_1 \times \mathbf{d}_2| = \sqrt{1 + 1 + 1} = \sqrt{3}$. Distance $= |-2|/\sqrt{3} = 2/\sqrt{3} = 2\sqrt{3}/3$.

13 [Intermediate] • Target: Foot of the perpendicular from a point to a line

Find the foot of the perpendicular from the point $P(4, 0, 1)$ to the line $\mathbf{r} = (1, 0, 0) + t(1, 1, 1)$.

A general point on the line is $(1 + t,\ t,\ t)$. The vector from this point to P is $(4 - (1 + t),\ 0 - t,\ 1 - t) = (3 - t,\ -t,\ 1 - t)$. For the foot of the perpendicular, this vector must be perpendicular to the direction $(1, 1, 1)$: $(3 - t)(1) + (-t)(1) + (1 - t)(1) = 0$, i.e. $3 - t - t + 1 - t = 0$, so $4 - 3t = 0$, giving $t = 4/3$. Foot of perpendicular: $(1 + 4/3,\ 4/3,\ 4/3) = (7/3, 4/3, 4/3)$.

14 [Intermediate] • Target: Foot of the perpendicular from a point to a line

Determine whether the planes $2x - y + 3z = 4$ and $-4x + 2y - 6z = -8$ represent the same plane, a pair of parallel (distinct) planes, or intersecting planes.

The second equation's coefficients $(-4, 2, -6)$ are exactly -2 times the first equation's coefficients $(2, -1, 3)$, and the constant -8 is also -2 times 4. Since the entire equation is a scalar multiple of the first, the two equations describe the same plane.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Determine whether the point $P(2, -1, 4)$ lies on the plane $3x + 2y - z = 1$, and if not, state whether it lies on the same side as the origin.

Substituting P: $3(2) + 2(-1) - 4 = 6 - 2 - 4 = 0 \neq 1$, so P does not lie on the plane. Substituting the origin $(0, 0, 0)$: $3(0) + 2(0) - 0 = 0$. Since the plane value at P (0) equals the plane value at the origin (0), and both are less than the required value of 1, P lies on the same side of the plane as the origin.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Find the value of k such that the line $\mathbf{r} = (1, 2, 0) + t(2, -1, k)$ is parallel to the plane $x + 2y - z = 5$.

A line is parallel to a plane when its direction vector is perpendicular to the plane's normal vector. Normal of the plane: $(1, 2, -1)$. Direction of line: $(2, -1, k)$. Setting the dot product to zero: $1(2) + 2(-1) + (-1)(k) = 0$, so $2 - 2 - k = 0$, giving $k = 0$.

17 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 3.01 3D Vectors & the Cross Product

A surveyor models a hillside as the plane $2x + 3y + 6z = 42$, where x and y are horizontal coordinates (in metres) and z is height (in metres). Find the height z of the hillside at the point where $x = 6$ and $y = 4$, and find the plane's intercept on the z-axis (i.e. the height at $x = 0, y = 0$).

Substituting $x = 6, y = 4$: $2(6) + 3(4) + 6z = 42$, so $12 + 12 + 6z = 42$, giving $6z = 18$, so $z = 3$ metres. The z-intercept (at $x = 0, y = 0$): $6z = 42$, so $z = 7$ metres.

18 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

This question has three parts. Two lines are given by $\mathbf{r}_1 = (2, 0, 1) + s(1, 1, 0)$ and $\mathbf{r}_2 = (0, 2, 1) + t(1, -1, 0)$. (a) Show that both lines lie in the plane $z = 1$. (b) Find the point of intersection of the two lines. (c) Find the acute angle between the two lines, to the nearest degree.

(a) Both lines have z-component fixed at 1 (since the direction vectors $(1, 1, 0)$ and $(1, -1, 0)$ have zero z-component, and both starting points have $z = 1$), so every point on either line satisfies $z = 1$; both lines lie in the plane $z = 1$. (b) Equating: $2 + s = t$ and $s = 2 - t$. Substituting the second into the first: $2 + (2 - t) = t$, so $4 - t = t$, giving $t = 2$, then $s = 0$. Point: $\mathbf{r}_1$ at $s = 0$ gives $(2, 0, 1)$; checking $\mathbf{r}_2$ at $t = 2$: $(0, 2, 1) + 2(1, -1, 0) = (2, 0, 1)$ ✓. Intersection point: $(2, 0, 1)$. (c).

19 [Challenging] • Target: Converting between vector and Cartesian forms of a line • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Write the parametric equations of the line $\mathbf{r} = (2, 1, -3) + t(1, -2, 4)$, and hence give its Cartesian (symmetric) form.

Parametric: $x = 2 + t$, $y = 1 - 2t$, $z = -3 + 4t$. Solving each for t: $t = x - 2$, $t = (1 - y)/2$, $t = (z + 3)/4$. Cartesian form: $(x - 2)/1 = (1 - y)/2 = (z + 3)/4$.

20 [Challenging] • Target: Determining if two planes are parallel • Cross-Lesson: 4.02 The Poisson Distribution

Determine whether the planes $x - 2y + 3z = 5$ and $2x - 4y + 6z = 1$ are parallel, and if so, whether they represent the same plane.

The second normal $(2, -4, 6)$ is exactly 2 times the first normal $(1, -2, 3)$, so the planes are parallel. If they were the same plane, the constant term would also scale by 2: $2 \times 5 = 10$. Since the actual constant is $1 \neq 10$, the planes are parallel but distinct.

© ClipSAT — clipsat.org | All Rights Reserved

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The vector equation of a line

Relative to a fixed origin O, the lines L_1 and L_2 have vector equations $L_1: \mathbf{r} = (1; 0; -1) + \lambda(1; 2; 2), \quad L_2: \mathbf{r} = (2; -4; 7) + \mu(2; 1; -2).$ The unique points P on L_1 and Q on L_2 are such that the line segment PQ is perpendicular to both L_1 and L_2. A point R lies on L_1 such that the area of triangle PQR is 18. Given that the z-coordinate of R is positive, what is the exact distance from the origin O to the midpoint of the line segment QR?

a. $3\sqrt{5}$ ✓

b. $\sqrt{13}$

c. $6\sqrt{2}$

d. $\sqrt{77}$

1. **Determine the positions of P and Q:** Let P on L_1 have coordinates $(1 + \lambda, 2\lambda, -1 + 2\lambda)$ and Q on L_2 have coordinates $(2 + 2\mu, -4 + \mu, 7 - 2\mu)$. The displacement vector $\vec{PQ}$ is: $\vec{PQ} = ((2 + 2\mu) - (1 + \lambda); (-4 + \mu) - 2\lambda; (7 - 2\mu) - (-1 + 2\lambda)) = (1 + 2\mu - \lambda; -4 + \mu - 2\lambda; 8 - 2\mu - 2\lambda).$ Since $PQ \perp L_1$.

22 [Foundational] • Target: Determining if two lines intersect

The line l_1 has vector equation $\mathbf{r} = (1; 5; 0) + \lambda(1; -2; 2)$ and the line l_2 has vector equation $\mathbf{r} = (1; p; 2) + \mu(2; 3; q)$ where p and q are constants, and $\lambda, \mu \in \mathbb{R}$. Given that the lines l_1 and l_2 are perpendicular and intersect at a common point, find the value of p.

a. -2 ✓

b. 2

c. -4

d. 4

We solve the problem in three steps: **Step 1: Determine q using perpendicularity** The direction vectors of l_1 and l_2 are $\mathbf{d}_1 = (1; -2; 2)$ and $\mathbf{d}_2 = (2; 3; q)$. Since $l_1 \perp l_2$, their scalar (dot) product must equal zero: $\mathbf{d}_1 \cdot \mathbf{d}_2 = (1)(2) + (-2)(3) + (2)(q) = 0\ 2 - 6 + 2q = 0 \Rightarrow 2q = 4 \Rightarrow q = 2.$ **Step 2: Find the parameters λ and μ at the intersection** Equating the coordinates of l_1 and l_2.

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: The equation of a plane

A straight line passing through the origin is tangent to both the curve C_1 with equation $y = e^{kx}$, where k is a positive constant, and the curve C_2 with equation $y = \ln x$. Determine the exact distance between the two points of contact.

a. $(e-1)\sqrt{e^2+1}$ ✓

b. $e\sqrt{e^2+1}$

c. $(e-1)\sqrt{e^2-1}$

d. $\sqrt{e^4-e^2+1}$

We synthesize differentiation with coordinate geometry to find the two points of contact. **Step 1: Point of tangency on $C_2 : y = \ln x$** Let $A(x_1, y_1)$ be the point of contact on C_2, so $y_1 = \ln x_1$. The derivative is $\frac{dy}{dx} = \frac{1}{x}$, so the gradient of the tangent at A is $\frac{1}{x_1}$. Since the tangent line passes through the origin $(0, 0)$, its gradient is also: $m = \frac{y_1 - 0}{x_1 - 0} = \frac{\ln x_1}{x_1}$ Equating the two expressions for the gradient.

24 [Foundational] • Target: Angle between a line and a plane

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. **1** ✓

d. 5/3

The point P is on the unit circle, so its coordinates are $(\cos\theta, \sin\theta)$. We are given $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. The point Q has coordinates (x, y) and its distance from the origin is 5. The angle between the positive x-axis and OQ is α. Thus, $x = 5\cos\alpha$ and $y = 5\sin\alpha$. We are asked to find the maximum possible value of the product of the x-coordinates of P and Q, which is $\cos\theta \cdot x$. Substituting the given values, we h.

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Distance from a point to a plane

Two particles, A and B, move in the xy-plane. Their position vectors at time t seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.

a. $\sqrt{26}$ ✓

b. $2\sqrt{13}$

c. $3\sqrt{10}$

d. $4\sqrt{5}$

The position vectors of particle A and particle B at time t are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$. The vector representing the displacement from particle B to particle A at time t is $\mathbf{d}(t) = \mathbf{r}_A(t) - \mathbf{r}_B(t) = (2t - (3 + 5t); (4t - 1) - (1 - 2t)) = (-3t - 3; 6t - 2)$. The distance between the particles at time t is the magnitude of this displacement vector.

26 [Foundational] • Target: Finding the angle between two planes

Consider two points in the Cartesian plane, $P = (2, 3)$ and $Q = (5, 7)$. What is the component form of the vector $\vec{PQ}$?

a. $\langle 3, 4 \rangle$ ✓

b. $\langle -3, -4 \rangle$

c. $\langle 7, 10 \rangle$

d. $\langle 1, 4 \rangle$

To find the component form of a vector $\vec{PQ}$ from point $P(x_1, y_1)$ to point $Q(x_2, y_2)$, we subtract the coordinates of the initial point P from the coordinates of the terminal point Q. The formula is $\vec{PQ} = \langle x_2 - x_1, y_2 - y_1 \rangle$. Given $P = (2, 3)$ and $Q = (5, 7)$, we have $x_1 = 2$, $y_1 = 3$, $x_2 = 5$, and $y_2 = 7$. Applying the formula: $\vec{PQ} = \langle 5 - 2, 7 - 3 \rangle$ $\vec{PQ} = \langle 3, 4 \rangle$ Therefore, the component form of the vector $\vec{PQ}$ is.

27 [Foundational] • Target: Finding the intersection line of two planes

Line of intersection of two non-parallel planes: direction vector $=$

a. sum of normals

b. **cross product of normals** ✓

c. difference of normals

d. dot product of normals

Intersection direction $= \mathbf{n}_1 \times \mathbf{n}_2$.

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Testing whether a line lies within a plane

Two vectors are anti-parallel if angle between them is:

a. 0

b. $\pi/4$

c. π ✓

d. $\pi/2$

Anti-parallel: $\theta = \pi$ (opposite directions).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Finding the point where a line meets a plane

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$ ✓

c. 1

d. $\frac{\sqrt{2}}{2}$

The equation of a unit circle centered at the origin is $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $y = \frac{\sqrt{3}}{2}$. Substituting this into the equation of the unit circle: $x^2 + (\frac{\sqrt{3}}{2})^2 = 1$ $x^2 + \frac{3}{4} = 1$ $x^2 = 1 - \frac{3}{4}$ $x^2 = \frac{1}{4}$ Since point P lies in the first quadrant, its x-coordinate must be positive. Therefore, $x = \sqrt{\frac{1}{4}} = \frac{1}{2}$. Alternatively, we are given that the angle θ formed by the ray from the or.

30 [Intermediate] • Target: Shortest distance between two skew lines

The locus $|z - (1)| = |z - (4 + 3i)|$ is the perpendicular bisector of the segment joining which two points?

a. $(4, 3)$ and $(1, 0)$

b. $(1, 0)$ and $(4, 3)$ ✓

c. $(-1, 0)$ and $(-4, -3)$

d. $(1, 3)$ and $(4, 0)$

Working through this additional-topics step: $|z - z_1| = |z - z_2|$ is the perpendicular bisector of the segment joining $z_1 = (1, 0)$ and $z_2 = (4, 3)$.

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Foot of the perpendicular from a point to a line

Consider a circle with the equation $x^2 + y^2 = 25$. A point $P(x_0, y_0)$ lies on this circle. A tangent line to the circle at point P intersects the x-axis at point A and the y-axis at point B. If the area of the triangle OAB, where O is the origin $(0, 0)$, is minimized, find the value of $x_0^2 + y_0^2$.

a. **25 ✓**

b. 50

c. 100

d. 125

The equation of the circle is $x^2 + y^2 = 25$. Let the point of tangency be $P(x_0, y_0)$. The equation of the tangent line to the circle at $P(x_0, y_0)$ is given by $xx_0 + yy_0 = 25$. To find the x-intercept A, we set $y = 0$: $xx_0 = 25 \Rightarrow x = \frac{25}{x_0}$. So, $A = (\frac{25}{x_0}, 0)$. To find the y-intercept B, we set $x = 0$: $yy_0 = 25 \Rightarrow y = \frac{25}{y_0}$. So, $B = (0, \frac{25}{y_0})$. The area of triangle OAB is.

32 [Intermediate] • Target: Checking whether a point lies on a plane

A rectangular garden is to be enclosed by a fence. One side of the garden is adjacent to a long straight river, and requires no fencing. The other three sides require fencing, and the total length of the fence available is 120 meters. Let the side of the garden parallel to the river have length x meters, and let the two sides perpendicular to the river each have length y meters. The area of the garden is given by $A(x, y) = xy$. \\ Find the dimensions x and y that maximize the area of the garden.

a. **x = 60 m, y = 30 m ✓**

b. x = 30 m, y = 60 m

c. x = 40 m, y = 40 m

d. x = 80 m, y = 20 m

The total length of the fence is 120 meters. The fence is used for the three sides of the rectangle not along the river. Thus, the constraint equation is $x + 2y = 120$. \\ We want to maximize the area $A = xy$. From the constraint equation, we can express x in terms of y: $x = 120 - 2y$. \\ Substitute this expression for x into the area formula: $A(y) = (120 - 2y)y = 120y - 2y^2$. \\ To find the maximum area, we need to find the critical points of $A(y)$ by taking the derivative with respect to y and setti.

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Converting between vector and Cartesian forms of a line

The diagram shows the graphs of the linear equation $y = 2x + c$ and the quadratic equation $y = -x^2 + 4x + 5$. The two graphs intersect at the points where $x = -1$ and $x = k$. Find the value of c and the value of k.

a. **c = 3, k = 4** ✓

b. c = 5, k = 3

c. c = 3, k = 3

d. c = 5, k = 4

The points of intersection satisfy both equations. Therefore, we can set the equations equal to each other: $2x + c = -x^2 + 4x + 5$ Rearranging this equation gives: $x^2 - 2x + (c - 5) = 0$ We are given that the graphs intersect at $x = -1$. Substituting $x = -1$ into the quadratic equation $y = -x^2 + 4x + 5$: $y = -(-1)^2 + 4(-1) + 5 = -1 - 4 + 5 = 0$ So, the point of intersection is $(-1, 0)$. Since this point also lies on the line $y = 2x + c$, we can substitute $x = -1$ and $y = 0$ into the linear equation.

34 [Intermediate] • Target: Determining if two planes are parallel

A particle of mass m kg is placed on a rough inclined plane. The plane is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{3}{4}$. The particle is on the point of sliding down the plane. The coefficient of static friction between the particle and the plane is μ_s. Given that the acceleration due to gravity is $g = 9.8$ m/s^2, find the value of μ_s.

a. $\frac{1}{2}$

b. $\frac{3}{4}$ ✓

c. $\frac{3}{5}$

d. $\frac{4}{5}$

When a particle is on the point of sliding down an inclined plane, the angle of inclination α is equal to the angle of friction ϕ, where $\tan\phi = \mu_s$. Given that $\tan\alpha = \frac{3}{4}$, and the particle is on the point of sliding, we have $\alpha = \phi$. Therefore, $\mu_s = \tan\phi = \tan\alpha = \frac{3}{4}$.

35 [Intermediate] • Target: Distance between two parallel planes

Distance between skew lines $L_1 : \mathbf{r} = \mathbf{a} + s\mathbf{d}_1$ and $L_2 : \mathbf{r} = \mathbf{b} + t\mathbf{d}_2$:

a. $|\mathbf{b} - \mathbf{a}|$

b. $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$ ✓

c. $|\mathbf{d}_1 - \mathbf{d}_2|$

d. 0

Applying Vectors and Vector-Valued Functions principles, Skew line distance formula. Therefore, the verified correct option is $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$.

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: Checking whether three points are collinear

A particle in equilibrium is acted on by three concurrent forces. Two of the forces have equal magnitude 20 N and act symmetrically either side of the third force's line of action. If the angle between each of the two equal forces and the third force is 120°, find the magnitude of the third force.

a. 18

b. **20 ✓**

c. 25

d. 22

By symmetry (each pair at 120°), the third force balances the resultant of the two equal forces: $F_3 = F_1 = 20$ N (standard 120°-120°-120° symmetric equilibrium).

37 [Intermediate] • Target: Reflecting a point in a plane

Two planes are parallel if their normal vectors are:

a. **parallel (scalar multiples) ✓**

b. perpendicular

c. equal in magnitude

d. zero

Parallel planes: normals are scalar multiples.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The vector equation of a line • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.

a. (1, 2)

b. **(2, 1) ✓**

c. (3, 0)

d. (0, 3)

The point of intersection of two lines is the solution to the system of simultaneous equations that represent these lines. By visually inspecting the graph, we can identify the coordinates where the two lines cross. The first line appears to pass through the points (0, 3) and (3, 0). The second line appears to pass through the points (0, 1) and (2, 3). Let's find the equations of the lines. For the first line: Gradient $m_1 = \frac{0-3}{3-0} = \frac{-3}{3} = -1$. Using the point-slope form with (3, 0).

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Determining if two lines intersect • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.

a. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((5 - x^2)^2 - r^2)dx$ ✓

b. Limits of integration are 0 and 4; $V_{\text{hole}} = \pi\int_0^4((4 - x^2)^2 - r^2)dx$

c. Limits of integration are 1 and 4; $V_{\text{hole}} = \pi\int_1^4((5 - x^2)^2 - r^2)dx$

d. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((4 - x^2)^2 - r^2)dx$

The region is bounded by $y = x^2$ and $y = 4$. To find the intersection points, we set $x^2 = 4$, which gives $x = \pm 2$. These are our limits of integration, $a = -2$ and $b = 2$. The region is revolved around the line $y = 5$. The outer radius $R(x)$ is the distance from the axis of revolution ($y = 5$) to the outer boundary of the region. The outer boundary is the curve $y = x^2$. So, $R(x) = 5 - x^2$. The inner radius of the cylindrical hole is given as r. The volume of the material removed by the cylindrical.

40 [Challenging] • Target: The equation of a plane • Cross-Lesson: 4.02 The Poisson Distribution

The half-line $\arg(z) = 45^\circ$ intersects the circle $|z| = 4$ at which point?

a. $(2\sqrt{2}3, 2\sqrt{2})$

b. $(-2\sqrt{2}, 2\sqrt{2})$

c. $(2\sqrt{2}, -2\sqrt{2})$

d. $(2\sqrt{2}, 2\sqrt{2})$ ✓

Working through this additional-topics step: $z = r(\cos 45^\circ + i\sin 45^\circ) = 4(\cos 45^\circ + i\sin 45^\circ) = (2\sqrt{2}, 2\sqrt{2})$

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Angle between a line and a plane • Cross-Lesson: 5.01 Further Integration Techniques

Consider a point $P(x, y)$ that lies on the parabola defined by the equation $y = x^2 - 6x + 13$. Let F be the focus of this parabola. If the distance from P to the directrix of the parabola is minimized, what is the y -coordinate of the point P?

a. 3

b. **4** ✓

c. 5

d. 6

The given parabola is $y = x^2 - 6x + 13$. We can rewrite this equation in vertex form by completing the square: $y = (x^2 - 6x + 9) + 13 - 9$, which simplifies to $y = (x - 3)^2 + 4$. This is the equation of a parabola with vertex $(h, k) = (3, 4)$, opening upwards. The standard form of a parabola opening upwards is $(x - h)^2 = 4p(y - k)$, where (h, k) is the vertex and p is the distance from the vertex to the focus and from the vertex to the directrix. Comparing our equation $(x - 3)^2 = 1(y - 4)$ with the s.

42 [Challenging] • Target: Distance from a point to a plane • Cross-Lesson: 5.02 Differential Equations

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

The problem states that point $P(x, y)$ lies on the unit circle, meaning $x^2 + y^2 = 1$. We are given that the y-coordinate of P is $y = \frac{3}{5}$. Substituting this into the unit circle equation, we get $x^2 + (\frac{3}{5})^2 = 1$. This simplifies to $x^2 + \frac{9}{25} = 1$. Subtracting $\frac{9}{25}$ from both sides gives $x^2 = 1 - \frac{9}{25} = \frac{16}{25}$. Taking the square root of both sides, we get $x = \pm\frac{4}{5}$. The problem also states that point P is in the second quadrant, where the x.

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • Target: Finding the angle between two planes • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

The graph of $y = f(x)$ is a curve passing through the points A(1, 2), B(4, 2), and C(2, 5). This graph is then subjected to a sequence of transformations. First, it is reflected in the line $y = x$. Let the image of the points A, B, and C under this reflection be A', B', and C' respectively. Secondly, the graph of the reflected curve is translated by the vector $(-3; 0)$. Let the final image of the points A', B', and C' after this translation be A'', B'', and C''. Given that the original curve is a parabola with its axis of symmetry parallel to the y-axis, and passes through A, B, and C, find the coordinates of C''.

a. (-3, 2)

b. (-2, -3)

c. **(2, -3)** ✓

d. (-3, -2)

The original points are A(1, 2), B(4, 2), and C(2, 5). The curve $y = f(x)$ is a parabola. Since A and B have the same y-coordinate, the axis of symmetry is $x = \frac{1+4}{2} = 2.5$. The equation of the parabola can be written as $y = a(x - 2.5)^2 + k$. Substituting C(2, 5): $5 = a(2 - 2.5)^2 + k \Rightarrow 5 = 0.25a + k$. Step 1: Reflection in the line $y = x$. This transformation swaps the x and y coordinates. A'(2, 1), B'(2, 4), C'(5, 2). Step 2: Translation by the vector $(-3; 0)$. This subtracts 3.

44 [Challenging] • Target: Finding the intersection line of two planes • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A particle's position in the xy-plane is given by the parametric equations $x(t) = 2t + 1$ and $y(t) = t^2 - 3$, where t is time. What is the Cartesian equation of the path traced by the particle?

a. $y = \frac{1}{2}x^2 - 3$

b. $y = \left(\frac{x-1}{2}\right)^2 - 3$ ✓

c. $y = x^2 - 3$

d. $y = 2\left(\frac{x-1}{2}\right) + 1$

To find the Cartesian equation, we need to eliminate the parameter t from the given parametric equations. Given: $x(t) = 2t + 1$ $y(t) = t^2 - 3$ From the first equation, we can solve for t in terms of x: $x - 1 = 2t$ $t = \frac{x-1}{2}$ Now, substitute this expression for t into the second equation: $y = \left(\frac{x-1}{2}\right)^2 - 3$ This is the Cartesian equation of the path traced by the particle.

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Testing whether a line lies within a plane • Cross-Lesson: 6.01 Matrices (AI HL)

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. What are the components of the vector $\mathbf{v}$?

a. ■**3, 4**■ ✓

b. ■7, 10■

c. ■-3, -4■

d. ■4, 3■

The components of a vector from point $P = (x_1, y_1)$ to point $Q = (x_2, y_2)$ are given by $\langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, $P = (2, 3)$ and $Q = (5, 7)$. Therefore, the components of vector $\mathbf{v}$ are $\langle 5 - 2, 7 - 3 \rangle = \langle 3, 4 \rangle$.

© ClipSAT — clipsat.org | All Rights Reserved